Jorge Sandoval

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Research Experience

2026 **EPSRC Postdoctoral Fellow, University of Dundee (UK)**: Research in Numerical Fluid Mechanics and Dynamical Systems in Wave-Current Boundary Layers.

2023-2025 **Postdoctoral Research Assistant, University of Dundee (UK)**: Research in Numerical Fluid Mechanics and Dynamical Systems as part of the EPSRC-funded project "Saving energy via drag reduction: a mathematical description of oscillatory flows".

2019 **Full-time Research Engineer Internship, University of Edinburgh (UK)**: Conceptualisation of numerical models to represent the interaction between waves and strong currents in tidal channels

2017-2019 **Research Engineer, Marine Research and Innovation Center (Chile)**: i) High-fidelity numerical and experimental analysis of marine hydrokinetic turbines. ii) Field measurements for hydrodynamic characterisation of the Magellan Strait and Fitz-Roy Channel.

Education

University of Edinburgh - Universidad Católica de Chile *Edinburgh, UK – Santiago, Chile*
PhD in Engineering (Joint Degree UoE-UC) *2019 – 2025*

- **Supervisors**: Professors David Ingram (UoE) and Cristián Escauriaza (UC)
- **Thesis title**: Unsteady three-dimensional flows dominated by free-surface dynamics: numerical insights for fluvial and coastal applications

Universidad Católica de Chile *Santiago, Chile*
MSc in Civil and Environmental Engineering *2015 – 2018*

- **Supervisors**: Professors Cristián Escauriaza (UC) and Emannuel Mignot (INSA-Lyon, France)
- **Thesis title**: Field and numerical investigation of transport mechanisms in a surface storage zone

Universidad Católica de Chile *Santiago, Chile*
Bachelor and professional degree in Civil Engineering *2009 – 2015*

- Specialisation in Hydraulic Engineering

Publications

Published

- **Sandoval, J.**, Eaves, T. Edge states and the periodic self-sustaining process in the Stokes boundary layer. *Journal of Fluid Mechanics*. 2025;1022:A10. doi:10.1017/jfm.2025.10725
- **Sandoval, J.**, Soto-Rivas, K., Gotelli, C., Escauriaza, C. (2021). Modeling the wake dynamics of a marine hydrokinetic turbine using different actuator representations. *Ocean Engineering*, 222 https://doi.org/10.1016/j.oceaneng.2021.108584
- Winckler, P., Molteni, F., Reyes, M., Gubler, A., **Sandoval, J.**, & Aleixo, R. (2021). Is Rhodamine a good tracer to predict coal transport in water?. *Obras y Proyectos*, (30), 16-29.
- **Sandoval, J.**., Mignot, E., Mao, L., Pastén, P., Bolster, D., Escauriaza, C. (2019). Field and Numerical Investigation of Transport Mechanisms in a Surface Storage Zone. *Journal of Geophysical Research: Earth Surface, 124*. https://doi.org/10.1029/ 2018JF004716
- Vargas, X. M., McPhee, J. T., Vicuña, S., Suarez, F., Meza, F., Daniele, L., Rondanelli, R., Lagos, M. Z., Mendoza, P., Bambach, N., Boisier, J. P., Cepeda, J. A., Vásquez, N. P., Morales, D., **Sandoval, J.**, Negri, A., & Caro, A. (2017). Actualización balance hídrico en Chile. Metodología y desafíos en modelación. *Proceedings of XXIII Congreso Chileno de Ingeniería Hidráulica*
- Gironás, J., **Sandoval, J.** (2017). Riesgo de origen hidrometeorológico en la ciudad de Santiago. *Santiago Resiliente*

In Preparation

- **Sandoval, J.**, D. , Eaves, T. Drag reduction in the asymptotic suction boundary layer via spanwise wall oscillations. In preparation for the *Journal of Fluid Mechanics*

Sandoval, J., D. , Eaves, T. Stokes boundary layer under the effect of a weak cross flow shear: edge states and coherent structure dynamics. In preparation for the *Journal of Fluid Mechanics*

- Gotelli., C, **Sandoval, J.**, Musa, M. Coupled interaction between fluvial dunes and Hydrokinetic Turbines. In preparation for *Renewable Energy*

- **Sandoval, J.**, Ingram, D. , Escauriaza, C. A single-phase Level-Set method for unsteady free surface flows in generalised curvilinear coordinates: geometric-based reinitialisation and physically-consistent dynamic boundary conditions. In preparation for the *Journal of Computational Physics*

Grants and Funded Projects

UKRI Mathematical Sciences Small Grant, UK Research and Innovation - (EPSRC)
Co-lead investigator. Funded original research in Wave-Current Boundary Layers. Awarded £100,000 over 12 months (2026). Competitive grant supporting mathematical sciences research

Conferences and Presentations

Invited Talks

Transition to turbulence in the Stokes Boundary Layer: Edge States and Periodic Self-Sustained Process (PSSP) **(Autumn 2025)** . Mathematics Seminar at the University of Dundee, UK.

Three-dimensional turbulent flows dominated by free-surface dynamics: new insights for fluvial and coastal applications in renewable energy **(Autumn 2023)**. Institute for Infrastructure and Environment seminar at the University of Edinburgh, UK.

Modelling of Hydrodynamic Processes in Aquatic Systems: From Contaminant Transport in Rivers to Marine Energy. **(Spring 2020)**. Civil Engineering Seminar at the University of Concepción, Chile.

Contributed Oral Presentations

European Fluid Dynamics Conference — *Dublin, Ireland, 2025*
Transition to turbulence in the Stokes Boundary Layer: Edge States and the Periodic Self-Sustained Process (PSSP)

European Fluid Dynamics Conference — *Dublin, Ireland, 2025*
Edge states in an oscillating boundary layer

International Couette-Taylor Workshop — *Durham, UK, 2025*
Transition to turbulence in the Stokes Boundary Layer: Edge States and the Periodic Self-Sustained Process (PSSP)

SIAM Conference on Applications of Dynamical Systems — *Denver, CO, USA, 2025*
Transition to turbulence in the Stokes Boundary Layer: Edge States and the Periodic Self-Sustained Process (PSSP) **(awarded with a travel fund by SIAM)**

Scottish Fluid Mechanics Meeting — *Glasgow, UK, 2025*
Transition to turbulence in the Stokes Boundary Layer: Edge States and the Periodic Self-Sustained Process (PSSP)

American Physical Society - Division of Fluid Dynamics Meeting — *Salt Lake City, UT, USA, 2024*
Transition to turbulence in the Stokes Boundary Layer: Edge States and Unsteady Self-Sustained Process (USSP)

Bifurcations and Instabilities in Fluid Dynamics — *Edinburgh, UK, 2024*
Transition to turbulence in the Stokes boundary layer. Part 2: Edge States

American Physical Society - Division of Fluid Dynamics Meeting — *Washington, DC, USA, 2023*
Vortical dynamics of supercritical flows in the vicinity of hydraulic structures

American Physical Society - Division of Fluid Dynamics Meeting — *Phoenix, AZ, USA, 2021*
Hydrodynamics of Mixing and Transport in Transient Storage Zones

American Physical Society - Division of Fluid Dynamics Meeting — *Phoenix, AZ, USA, 2021*
Turbulent transport dynamics in open-channel floodplains for different submergence conditions

American Physical Society - Division of Fluid Dynamics Meeting — *Seattle, WA, USA, 2020*
Wake dynamics in horizontal hydrokinetic tidal turbines using three coupled DES-Actuator Model approaches

River Flow meeting, IAHR 9th international conference on fluvial hydraulics *Lyon, France, 2018*
Turbulent flow dynamics and mass transport processes in a natural storage zone using field data and numerical simulations

American Physical Society - Division of Fluid Dynamics Meeting *Portland, OR, USA, 2016*
Turbulent Coherent-Structure Dynamics in a Natural Surface-Storage Zone **(awarded with a travel fund by APS)**

Poster Presentations

Scottish Fluid Mechanics Meeting *Edinburgh, UK, 2024*
A single-phase Level-Set method for unsteady free surface flows: geometric-based reinitialisation and physically-consistent dynamic boundary conditions

American Geophysical Union (AGU) Fall Meeting *Washington D.C., USA, 2018*
A coupled DES - Actuator Lines Model to represent the interaction between tidal flows and Marine Hydrokinetic Turbines

American Geophysical Union (AGU) Fall Meeting *San Francisco, CA, USA, 2015*
Hydrodynamic characterization of a surface storage zone in a natural stream

Teaching Experience

Lecturer

2021	**Fluid Mechanics**	*Universidad Diego Portales*

Teaching Assistant

2023-2024	**Partial Differential Equations III**	*University of Edinburgh*
2022-2023	**Computational Fluid Dynamics V**	*University of Edinburgh*
2022-2023	**Marine Renewable Resource Assessment**	*University of Edinburgh*
2022-2023	**Fluid Mechanics IV**	*University of Edinburgh*
2015-2017	**Fluvial Hydraulics**	*Universidad Católica de Chile*
2014	**Hydrological Modelling**	*Universidad Católica de Chile*
2014-2016	**Hydraulic Design and Analysis**	*Universidad Católica de Chile*
2013-2016	**Fluid Mechanics**	*Universidad Católica de Chile*

Workshops and Courses

CISM-ECCOMAS Advanced course on "Machine Learning for Fluid Mechanics" *Udine, Italy, 2023*
Introduction to advanced machine learning methods and their application to complex fluid mechanics problems, including modelling, control, turbulence closures, and shape optimisation.
Coordinators: Professor **Bernd R. Noack** (Harbin Institute of Technology Shenzhen, China) and Professor **Steven Brunton** (University of Washington, USA)

CISM-ECCOMAS International Summer School on "Coherent Structures in Unsteady Flows: Mathematical and Computational Methods" *Udine, Italy, 2019*
Introduction to coherent structures and their modern computational tools.
Organiser: Professor **George Haller**, ETH Zurich, Switzerland

Summer Institute on Earth-Surface Dynamics *Minneapolis, MN, USA, 2018*
Mathematical tools for Earth casting, focusing on landscape connectivity and how signals (e.g. climate change) propagate through and are recorded in Earth-surface systems.
Organiser: Professors **Chris Paola** and **Vaughan Voller**, University of Minnesota, USA

Other Professional Experience

2017 **Project Engineer, UC Climate Change Center (Chile)** – Hydrological modelling of surface processes for the National Water Balance Project using VIC

2015–2016 **Project Engineer, Hydraulic Models Laboratory, DICTUC** – Construction, testing, and operation of physical models of hydraulic structures

2016 **Project Engineer, Research Centre for Integrated Disaster Risk Management** – Numerical modelling of flood events in urban areas

Professional Skills

Language

English (proficient) and **Spanish** (native)

Computer Skills

Advance knowledge: Fortran, C++, Python, Matlab, OpenFOAM, HEC-RAS, MPI, Tecplot and Pointwise

Basic knowledge: Star-CCM+, CAD environment, Iber, VIC hydrological model

Awards

Award for excellence in teaching assistant career - School of Engineering at Universidad Católica de Chile (2015)

Padre Hurtado Scholarship - Universidad Católica de Chile (2009)

Referees

Tom Eaves (teaves001@dundee.ac.uk). Lecturer at the School of Science and Engineering, University of Dundee **(PI, Postdoctoral Research Assistant position project)**

David Ingram (david.ingram@ed.ac.uk). Professor of Computational Fluid Dynamics at School of Engineering, University of Edinburgh **(PhD supervisor)**

Cristián Escauriaza (cescauri@ing.puc.cl). Associate professor at Hydraulic and Environmental Engineering Department, Universidad Católica de Chile **(PhD and MSc supervisor)**